

SANKET v1.0 — Source Code Reproduction Guide

Complete Technical Instructions for Building, Testing, and Packaging SANKET from Source

System Name:	SANKET (AI-Assisted FSOC Workstation)	Target Organization:	Department of Space / ISRO
Problem Statement:	SIH 2026 PS 26169 (PS-4)	Clean Archive:	SANKET_Source.zip (30.36 MB, 281 files)
Backend Stack:	Python 3.11.9, OpenCV 4.10, PySide6	Frontend Stack:	React 19, TypeScript, Three.js, Vite

1. System Architecture & Module Organization

```
SANKET/
|-- src/
|   |-- aiml/           # Calibrated Logistic Regression classifier & feature extraction
|   |-- app/            # AppController & QtWebEngine host window
|   |-- control/        # PTZ dual-axis PID controller with rate limiting (<= 10 deg/s)
|   |-- evaluation/     # BenchmarkManager (19-scenario matrix & MP4 video evaluator)
|   |-- frame/          # SimulationFrameProvider and MP4FrameProvider
|   |-- metrics/        # Real-time MetricsEngine and LoggingEngine
|   |-- simulation/     # 2000x2000 World Canvas, FPA Camera Model, DisturbanceEngine
|   |-- tests/          # 599 automated unit, integration, and contract tests
|   |-- tracker/        # DetectionEngine, Sub-pixel CoG Centroid, Kalman State Machine
|-- frontend/           # Embedded React 19 / TypeScript / Tailwind / Three.js UI
|-- scenarios/          # 19 standardized ISRO PS-26169 scenario JSON definitions
|-- scripts/            # Screenshot capture, PDF compilers, and packaging scripts
-- sanket.spec          # PyInstaller standalone executable specification
```

2. Reproduction & Build Commands

Step 1: Environment Setup & Python Dependencies

```
python -m venv .venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

Step 2: Frontend Bundle Compilation

```
cd frontend
npm install
npm run build
cd ..
```

Step 3: Verification via Automated Test Suite (599 Tests)

```
pytest
```

Step 4: Launching SANKET

```
# Graphical Interactive Workstation:
python -m src.main

# Headless Batch Execution:
python -m src.main --headless --scenario scenario_2_circular --duration 30
```

Step 5: Standalone Binary Compilation (PyInstaller)

```
pyinstaller sanket.spec --noconfirm --clean
```

End of Source Code Reproduction Guide — SANKET v1.0 Production Release